When karyotype still matters
A published Indian family: two related children with opposite imbalances, and a carrier that neither karyotype nor CMA could see

Summary for doctors and healthcare teams, written in our own words from a published, peer-reviewed case report. Please read the original article (link below).

Source: Tuteja M, Agarwal D, Phadke SR. *Double Segment Chromosomal Imbalance due to Inherited Chromosomal Translocation: Detection by Cytogenetic Microarray*. Indian Pediatrics 2017;54:879-881. Department of Medical Genetics, Sanjay Gandhi Post Graduate Institute of Medical Sciences (SGPGIMS), Lucknow. indianpediatrics.net/oct2017/oct-879-881.htm

Patient 1 (the proband)

- Girl, referred at 5 months: atrial septal defect and patent ductus arteriosus, facial dysmorphism
- Broad thumbs and great toes, camptodactyly, single palmar crease; growth below normal
- Delayed milestones; intellectual disability on review at 5 years
- G-banded karyotype (550 bands) and brain MRI: **normal**
- CMA: **gain at 7q36.1-q36.3 (about 9.4 Mb)** and **loss at 11q24.1-q25 (about 13.2 Mb)**

Patient 2 (a relative)

- Girl; her father is a cousin of patient 1's father
- Severe growth restriction, microcephaly, facial dysmorphism; agenesis of the corpus callosum on MRI; died at 8 months
- Prenatal karyotype (550 bands): **normal**
- CMA: the **opposite pattern**: loss at 7q36 and gain at 11q24-q25
- Pedigree: further affected pregnancies, including holoprosencephaly

How the carriers were found

Terminal imbalances of two different chromosomes, in opposite directions, in two related children point to a **balanced translocation** segregating in the family. A balanced carrier has no net loss or gain, so **CMA cannot detect it**, and here the exchanged segments were too small to see on routine G-banding (a **cryptic** translocation). **FISH** with centromeric probes for chromosomes 7 and 11 and subtelomeric probes for 7q36 and 11q24.1-q25 in patient 1's father **confirmed a 7;11 translocation**; two other male relatives were judged likely carriers. In a later pregnancy of patient 2's mother, **CVS at 11 weeks** (MLPA) showed the same imbalance, and the family was counselled. The authors stress a detailed family history, pedigree analysis and the use of microarray. The genes in the regions include **SHH** (7q36) and **CDON** (11q24), both linked to holoprosencephaly.

Talking points for the doctor visit

- CMA tells how much DNA is lost or gained; karyotype and FISH show how chromosomes are arranged. A balanced carrier needs karyotype or FISH.
- Opposite terminal imbalances on two chromosomes in a child: check the parents for a balanced translocation.
- Once the family rearrangement is known, offer carrier testing to relatives and prenatal testing in future pregnancies.
- Prenatal testing in India: written consent under the PCPNDT Act; disease testing only; fetal sex never disclosed.

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When to order a karyotype in the microarray era

Situation	Why karyotype (or FISH)
Down syndrome	CMA or QF-PCR shows extra chromosome 21 material, but only karyotype shows whether it is a free trisomy or a Robertsonian translocation. If a translocation is found, karyotype both parents. Quoted recurrence risks: about 10-15% if the mother is a balanced carrier, about 1-3% if the father is; almost 100% for a parental rob(21;21). India (AMPATH, Hyderabad): 1,597 Down syndrome cases among 9,313 karyotyped samples; free trisomy 91.7%; Robertsonian 90 (5.6%), most often rob(14;21) (42) and rob(21;21) (36).
Recurrent pregnancy loss	In 2-5% of couples, one partner carries a balanced rearrangement; carriers are healthy and CMA is normal. ASRM (2012): karyotype both parents. RCOG (2011): parental karyotype if the pregnancy tissue shows an unbalanced structural abnormality. ESHRE (2017): not routine; after individual assessment.
Turner syndrome	Short stature or primary amenorrhoea in girls. Count at least 30 cells so mosaicism is not missed (ACMG).
Male infertility	Primary infertility with azoospermia, or sperm concentration below 5 million/mL with raised FSH, testicular atrophy or impaired sperm production: karyotype (e.g. Klinefelter 47,XXY) (AUA/ASRM 2024, statement 12).
Mosaicism and markers	Karyotype examines individual cells; low-level mosaicism may be missed by CMA. A small supernumerary marker chromosome needs FISH or CMA to identify its origin.
Limits of karyotype	Changes smaller than about 5-10 Mb are not seen; use CMA for small deletions and duplications, FISH for a targeted region or carrier check. Often the tests work together.
Sample	Sodium heparin blood (green top); live cells are cultured, so do not freeze. Prenatal: amniotic fluid or CVS.

References

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- Pre-Conception and Pre-Natal Diagnostic Techniques (PCPNDT) Act, 1994 (India).

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